Installation and Configuration of Virtualization using KVM

```
pict@pict-OptiPlex-3020:~$ egrep -c '(vmx|svm)' /proc/cpuinfo
8
```

Step 1: Verify Hardware Virtualization Support

- **If output > 0 → virtualization supported**
- **If 0 → enable *Intel VT-x / AMD-V* in BIOS**

```
pict@pict-OptiPlex-3020:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for pict:
Get:1 https://packages.microsoft.com/repos/code stable InRelease [3,590 B]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 https://packages.microsoft.com/repos/code stable/main arm64 Packages [25.3 kB]
Get:6 https://packages.microsoft.com/repos/code stable/main amd64 Packages [25.2 kB]
Get:7 https://packages.microsoft.com/repos/code stable/main armhf Packages [25.4 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,576 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:14 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,380 B]
Get:15 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:16 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.2 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [252 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:21 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
```

Step 2: Update System

```
pict@pict-OptiPlex-3020:~$ sudo apt install -y qemu-kvm libvirt-daemon-system libvirt-clients bridge-utils virt-manager
[sudo] password for pict:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'qemu-system-x86' instead of 'qemu-kvm'
The following packages were automatically installed and are no longer required:
```

Step 3: Install KVM and Required Packages

```
pict@pict-OptiPlex-3020:~$ sudo systemctl status libvirtd
● libvirtd.service - libvirt legacy monolithic daemon
     Loaded: loaded (/usr/lib/systemd/system/libvirtd.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-04-08 09:42:34 IST; 1min 56s ago
TriggeredBy: ● libvirtd-admin.socket
             ● libvirtd-ro.socket
             ● libvirtd.socket
       Docs: man:libvirtd(8)
             https://libvirt.org/
   Main PID: 33215 (libvirtd)
      Tasks: 22 (limit: 32768)
     Memory: 11.4M (peak: 12.7M)
        CPU: 502ms
     CGroup: /system.slice/libvirtd.service
             ├─33215 /usr/sbin/libvirtd --timeout 120
             ├─33504 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default.conf
             └─33505 /usr/sbin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default.conf
```

Step 4: Check KVM Installation

Installation and Configuration of Virtualization using KVM

```
pict@pict-OptiPlex-3020:~$ kvm-ok
INFO: /dev/kvm exists
KVM acceleration can be used
```

Step 5: Also verify using command: kvm-ok

```
pict@pict-OptiPlex-3020:~$ sudo usermod -aG libvirt $USER
sudo usermod -aG kvm $USER
```

Step 6: Add User to Required Groups. Then log out and log back in/ reboot.

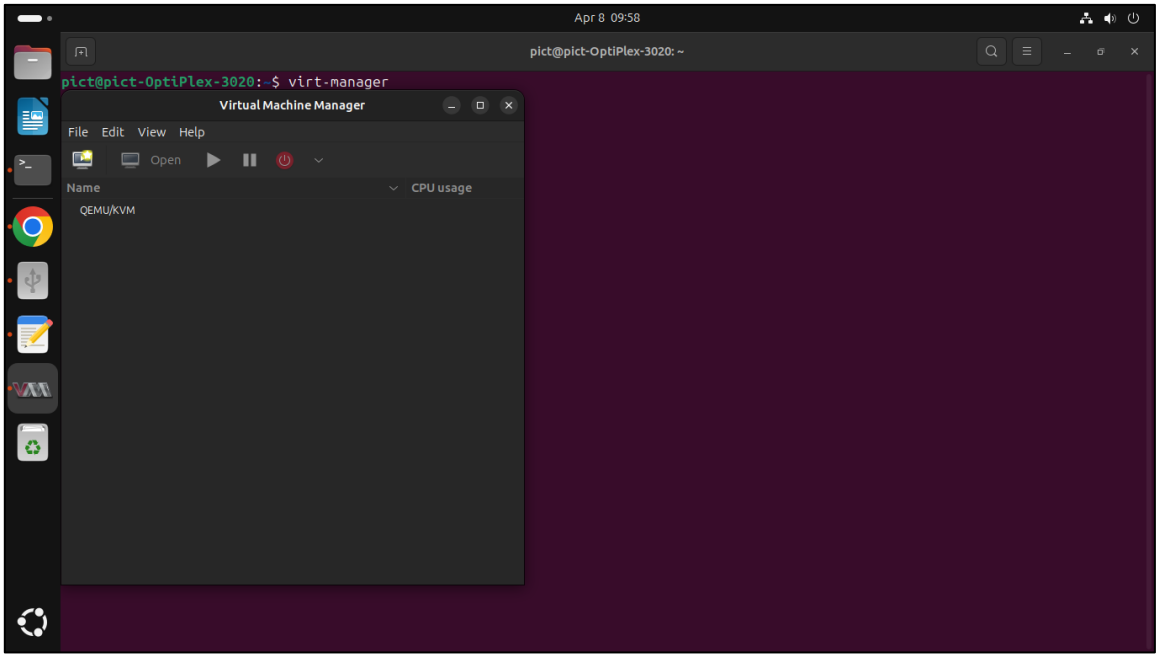
```
pict@pict-OptiPlex-3020:~$ groups
pict adm cdrom sudo dip plugdev users kvm lpadmin libvirt
```

Step 7: kvm added under groups

```
pict@pict-OptiPlex-3020:~$ virsh list --all
 Id   Name   State
--------------------
```

```
pict@pict-OptiPlex-3020:~$ virsh net-list --all
 Name      State    Autostart   Persistent
--------------------------------------------
 default   active   yes         yes
```

Step 8: Verify Setup

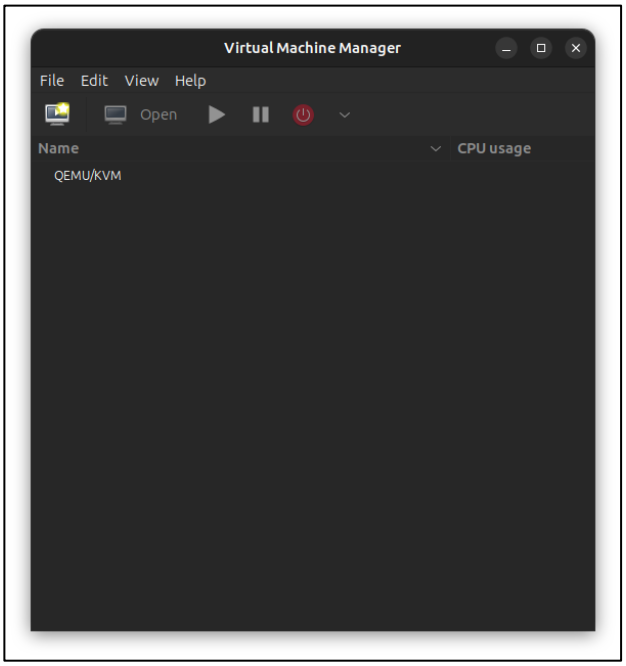


Step 9: Create Virtual Machine (GUI Method) using virt-manager command

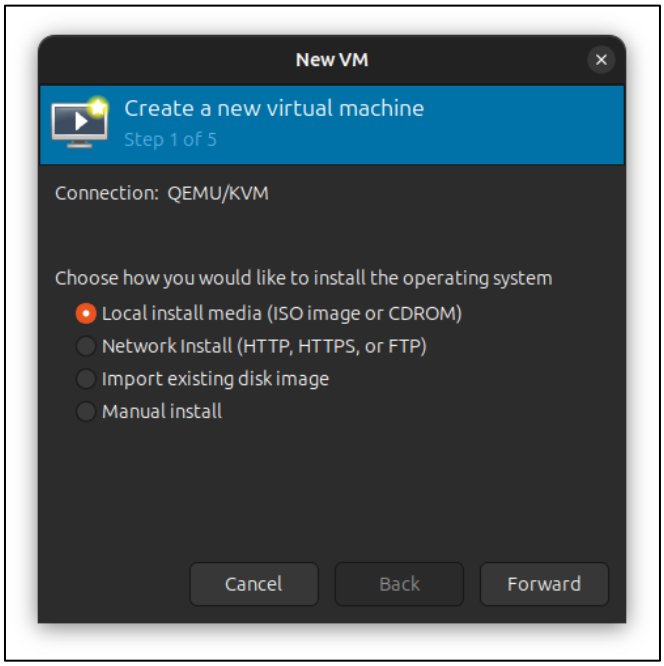
Roll No: 31446

Installation and Configuration of Virtualization using KVM

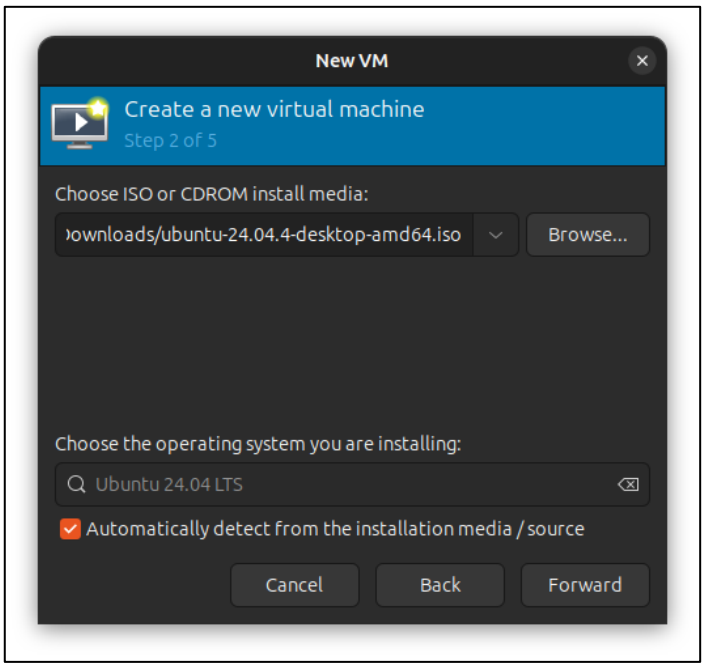
KEEP ISO FILE READY BY DOWNLOADING ANY ISO FILE OF AN OS (UBUNTU OS)



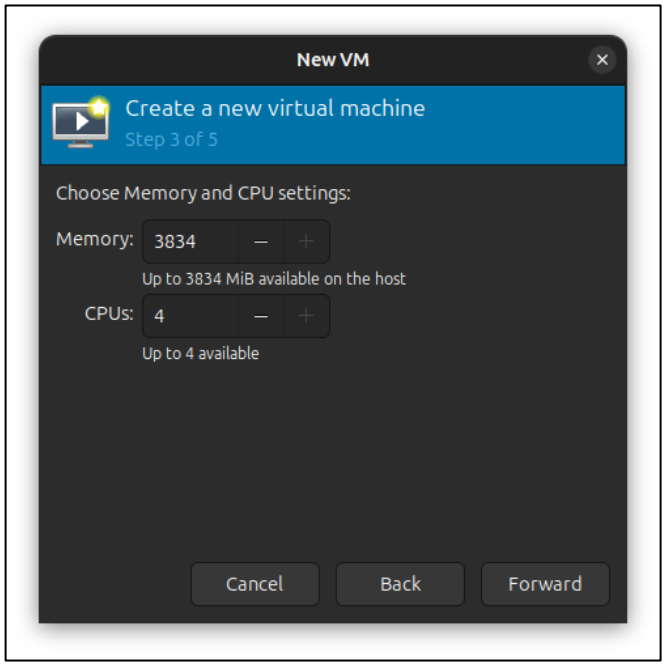
Step 10: Click File > New VM



Step 11: Choose first option

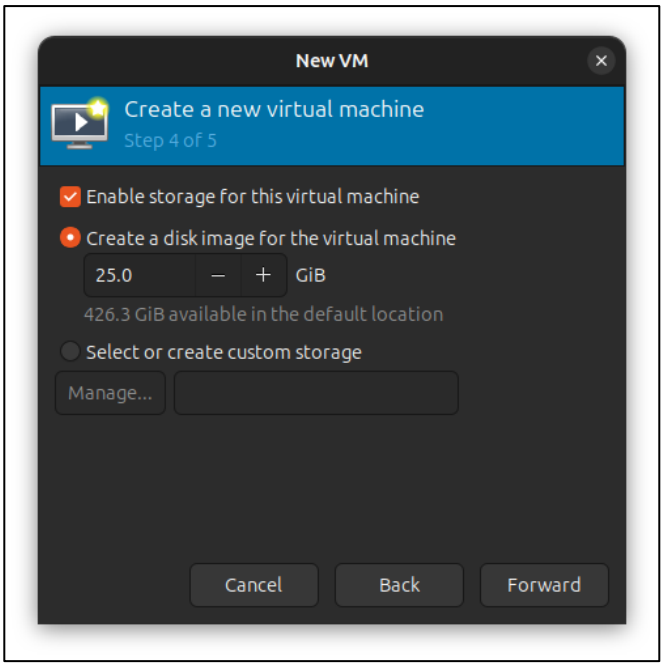


Step 12: Browse iso file (Ubuntu OS iso file)

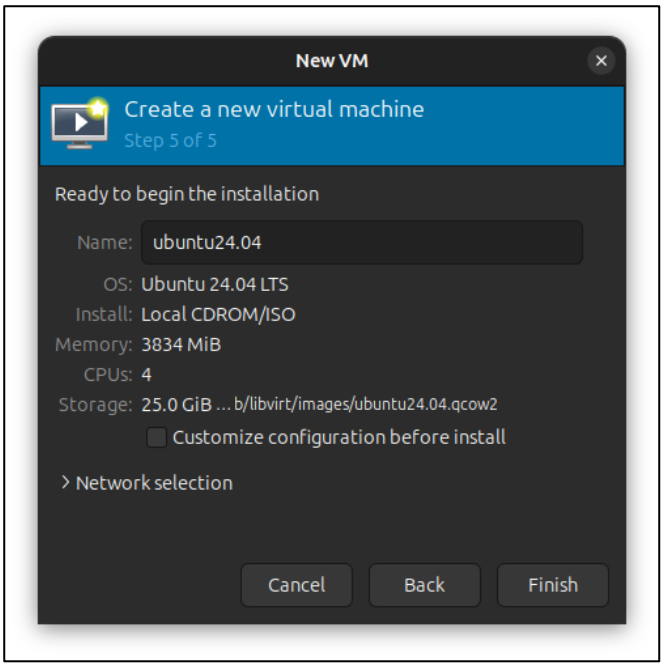


Step 13: Allocate Memory and CPU

Installation and Configuration of Virtualization using KVM



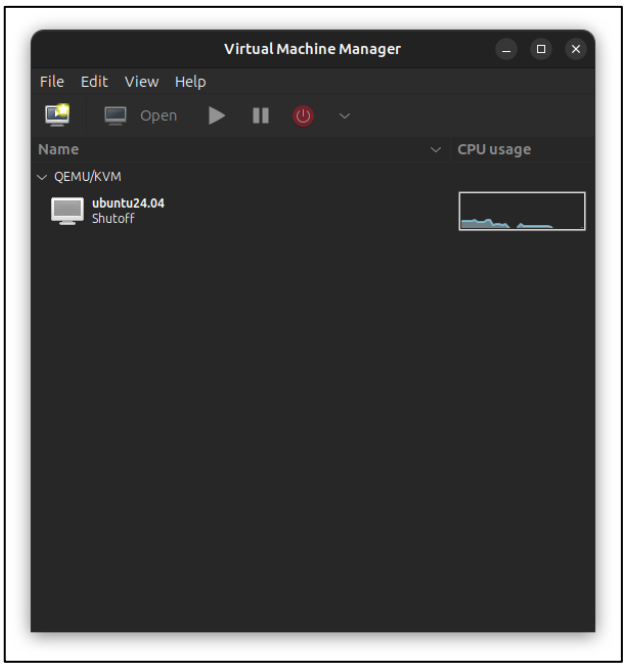
Step 14: Allocate Disk space for VM



Step 15: Review final details & finish

```
pict@pict-OptiPlex-3020:~$ virsh list
 Id   Name          State
----------------------------
 1    ubuntu24.04   running
```

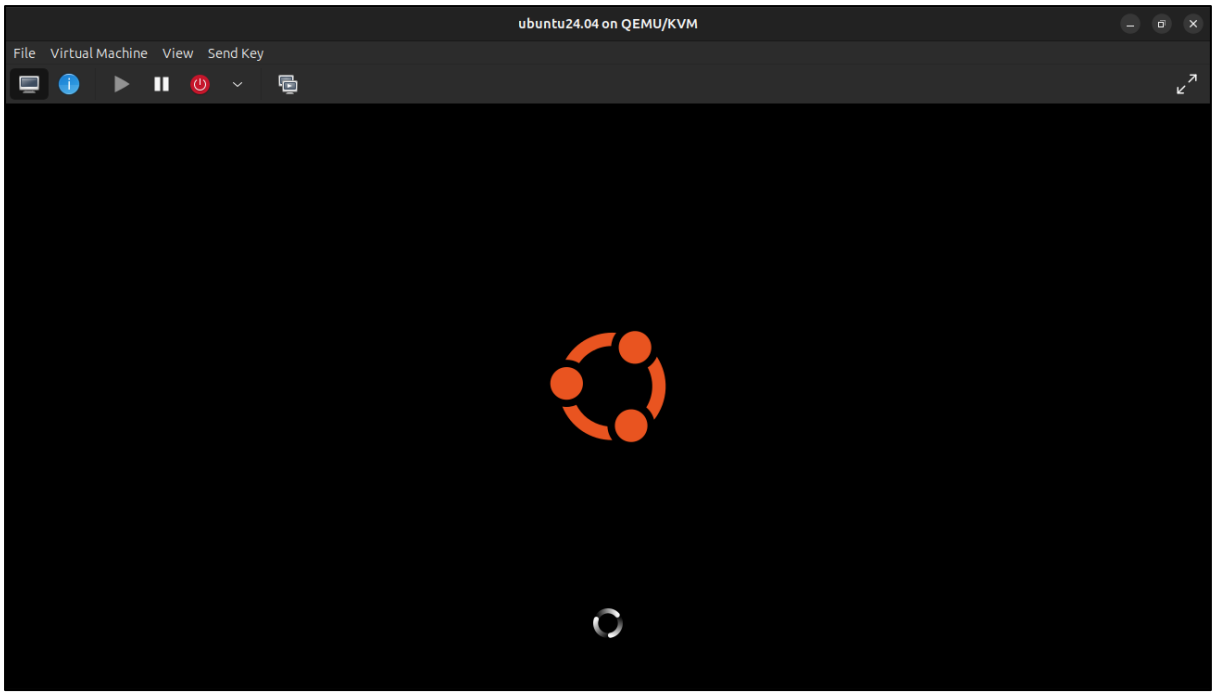
Step 16: Verify VM is running



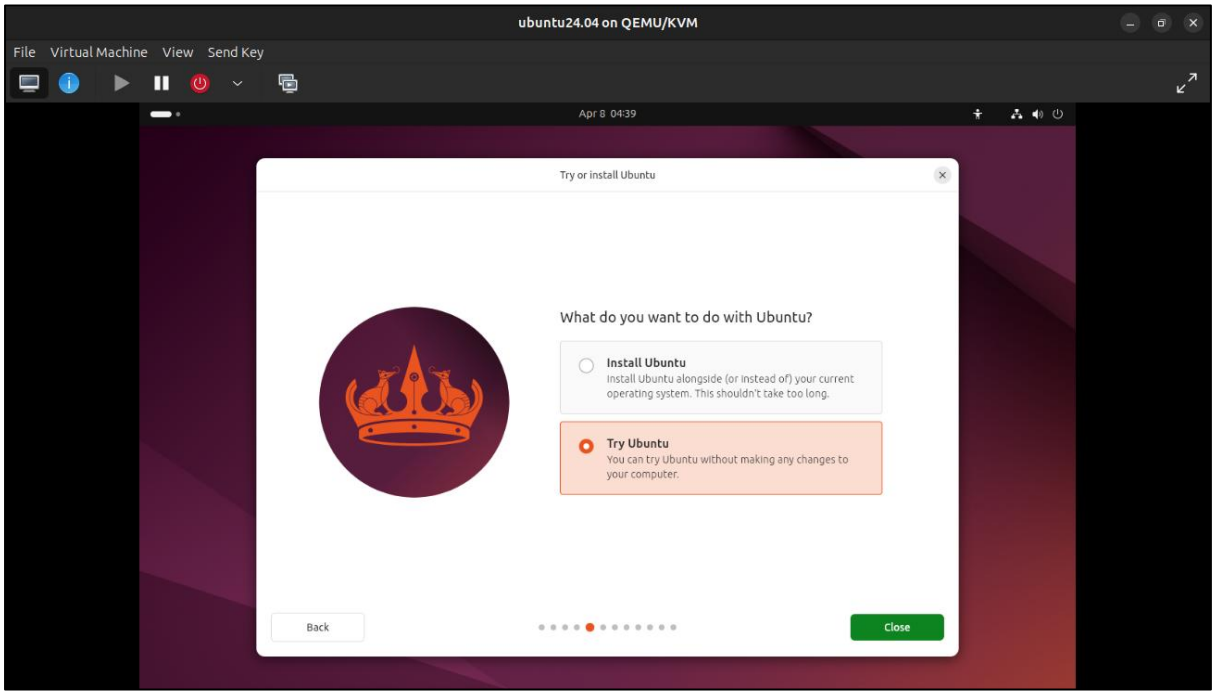
Step 17: Verify in virt-manager

Roll No: 31446

Installation and Configuration of Virtualization using KVM

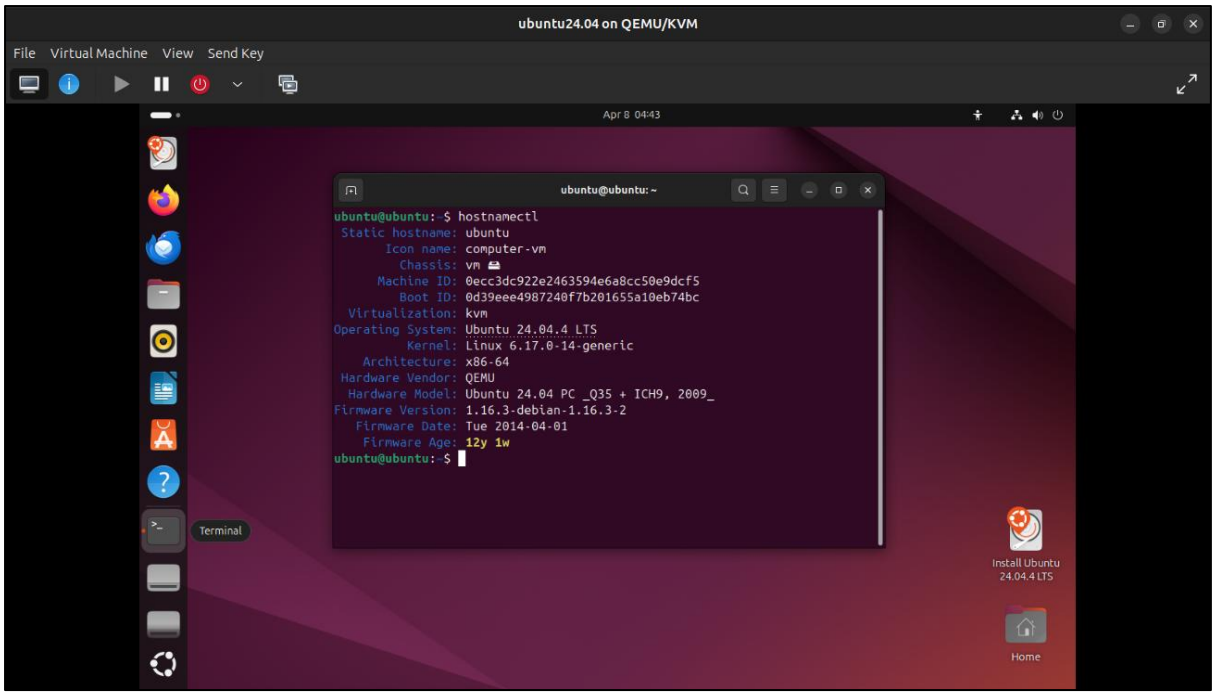


Step 18: Ubuntu will boot up



Step 19: Choose Try Ubuntu

Installation and Configuration of Virtualization using KVM



Step 20: Use command hostnamectl to check host details indicating you are on the VM